

Seed and Event Sensitivity in Lightweight SAR Flood Change Detection: A Controlled Negative Result and an Accuracy–Cost Benchmark

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Abstract. Lightweight change detectors are attractive for mapping floods in synthetic aperture radar (SAR) imagery, and small-object losses or boundary heads are natural additions when small urban floods matter. We report what happened when such additions were tested under pre-specified protocols instead of a single training run. On the UrbanSARFloods dataset we train a 1.1M-parameter Siamese network in a 2×2 factorial (size-weighted loss, boundary package) together with two baselines, FC-Siam-Diff and a SAR adaptation of BIT, over six seeds and two training recipes, and evaluate all 36 runs once on a sealed test set. Neither component has a consistent effect on validation or test, and every model recovers only 2–7% of interior small floods on test. The ranking between model families is stable across seeds, whereas the ranking among the factorial variants is not. Seed-to-seed variation concentrates in a single event on each split, but a pre-specified explanation based on flood fraction is not supported by the test set. A cost study shows that parameter count, counted FLOPs, GPU latency and CPU latency rank the models in four different orders. We conclude with reporting recommendations for this task.

Keywords: SAR · flood mapping · change detection · lightweight models · seed sensitivity · negative result · efficiency

1 Introduction

Synthetic aperture radar (SAR) observes the ground through cloud and at night, which makes it the usual data source for mapping floods while they happen [24]. Public datasets such as Sen1Floods11 [1] have made deep segmentation models the standard approach for open-area flooding. Urban flooding is harder. Buildings produce strong and geometry-dependent backscatter, flooded streets are narrow, and the flooded regions are often only a few pixels wide. The authors of UrbanSARFloods report that common training choices struggle with this imbalance and that urban flood detection remains challenging [24]; earlier work found that interferometric coherence helps in built-up areas [14].

Two further pressures shape model design for this task. First, flood maps are wanted quickly and, increasingly, close to the sensor, which favours small

models [16, 6]. Second, the regions that matter most to responders are small, so it is tempting to add components that target them: losses that up-weight small connected components [13, 18] and heads or losses that sharpen boundaries [12].

This paper started from exactly that design: a 1.1M-parameter Siamese network with a size-weighted Tversky loss and a thin boundary head. A single-seed pilot gave results that could be read as a trade-off between the two components. When we repeated the comparison under protocols fixed in advance, with new seeds, a second training recipe and finally a sealed test set, the component effects did not replicate. We therefore report the study as it is: a controlled negative result, together with what it shows about seed and event sensitivity and about the cost of the models involved.

We ask three questions. **RQ1**: do size weighting and a boundary package have a consistent effect across seeds, two training recipes and a held-out test set? **RQ2**: how stable are method rankings against seed and event variation? **RQ3**: what is the accuracy–cost trade-off of sub-5M-parameter change detectors on a GPU and on a CPU?

Our contributions are findings rather than a new method:

- A controlled negative result for both components across six seeds, two recipes and one sealed test evaluation (Sect. 4.1).
- An analysis of seed and event sensitivity, including a pre-specified explanation that the test set did not support (Sects. 4.2–4.4).
- An accuracy–cost benchmark of three model families with several cost indicators measured on two devices (Sect. 4.5).
- Protocols, decision rules, code and per-run records that allow each number to be traced.

Single-run comparisons are known to be fragile in machine learning [19, 3, 17], and simple baselines often remain competitive in change detection [7]. The present study adds a worked case for SAR flood mapping, where validation sets contain only a handful of events.

2 Related Work

SAR flood datasets. Sen1Floods11 provides georeferenced Sentinel-1 data for 11 flood events [1]. Kuro Siwo covers 43 events and provides both ground-range and single-look complex products [2]. UrbanSARFloods contains 8,879 chips of 512×512 pixels from 18 events, with pre- and co-event Sentinel-1 intensity and coherence and separate labels for flooded open areas and flooded urban areas [24]. We use its intensity channels only.

Bitemporal change detection. Flood mapping from a pre-event and a co-event image is a change-detection problem. Fully convolutional Siamese networks [8] and transformer-based models such as BIT [5] are common starting points, and Siamese designs with attention have been applied to SAR flood pairs [21, 20].

Two recent studies caution that architectural novelty explains less than it appears to: a plain U-Net remains a top performer on standard optical benchmarks [7], and a broad benchmark finds Siamese U-Nets competitive once efficiency is counted [23].

Small regions and boundaries. The Tversky loss trades false positives against false negatives [22]. Instance-aware losses weight each connected component rather than each pixel [13, 18], and boundary losses address highly unbalanced segmentation [12]. These ideas were developed mainly for medical images; small changed regions have also been studied in optical remote sensing [4]. Evaluation guidelines recommend object-level metrics when objects of very different sizes matter [15].

Efficiency. Lightweight change detectors report strong accuracy with few parameters [6], including for SAR flood mapping [11], and flood segmentation has been developed for use on board small satellites [16]. Dehghani et al. show that cost indicators such as parameters, FLOPs and speed can disagree and argue for reporting several of them [9].

Variance across runs. Reimers and Gurevych show that the random seed alone can change conclusions for sequence taggers [19]; Bouthillier et al. quantify several sources of variance in benchmarks [3]; and Picard documents seed effects in computer vision [17].

Gap. Our literature search was bounded (one search engine, 31 queries on 3 October 2026 and follow-up queries on 10 October 2026) and mostly read at abstract level. Within those searches we found no SAR flood study that reports recall for small flooded regions inside a patch, or that reports how method rankings change across training seeds. This is a statement about our searches, not a claim that no such study exists.

3 Experimental Design

3.1 Data and Splits

We use the Sentinel-1 intensity channels of UrbanSARFloods [24]: VH and VV backscatter in decibels for the pre-event and co-event dates. The two flood classes are merged into one binary flood class. Scenes are cut into non-overlapping 256×256 patches, and each input channel is standardized with statistics fitted on training pixels.

Events, not patches, are held out (Table 1). The dataset’s test events stay as test; the remaining events are re-partitioned into 12 training and 3 validation events by a fixed hash rule. *This split differs from the dataset’s official partition, so our numbers are not comparable with those in the dataset paper.* A connected component is *small* if its area within the patch is at most 35 pixels, the 25th

Table 1. Event-held-out split. “Positive” is the share of valid pixels labelled as flood. The training positive share was not recorded in our result records.

Split	Events	Patches	Positive	Components	Interior	small
Train	12	19,166	–	52,516	–	–
Validation	3	4,767	1.08%	6,745	370	370
Test	3	9,204	1.22%	27,175	4,899	4,899

percentile of training components, and *interior* if it does not touch the patch border or an invalid pixel.

The validation events are Lumberton (2016), Coraki (2022) and Hebei (2023). The test events are Weihui (2021; 1.03% positive), Nova Kakhovka (2023; 1.86%) and Jubba (2023; 0.85%). Nova Kakhovka contains 4,379 of the 4,899 interior small components of the test set.

3.2 Models

SmallFlood-CDNet is a Siamese network with a MobileNetV3-small encoder [10] trained from scratch, a temporal-difference fusion and a light decoder (1.08M parameters). Its loss is $0.4 \text{ BCE} + 0.4 \text{ Tversky}$ with false-positive and false-negative weights 0.3 and 0.7 [22]. We study two optional components in a 2×2 factorial:

- **Size weighting.** Positive pixels of a component with area a receive weight $1 + \min(4, (65/a)^{0.5})$ in both loss terms, where 65 pixels is the median training component area.
- **Boundary package.** A thin boundary head whose output is added to the change logits as a residual, with an auxiliary boundary loss of weight 0.2.

Variant A uses neither, B size weighting, C the boundary package and D both.

The baselines are *FC-Siam-Diff* [8] (0.49M parameters) and *BIT-SAR v2* (3.49M parameters), our two-channel adaptation of BIT [5] trained from scratch. Both use $0.5 \text{ BCE} + 0.5 \text{ Tversky}$. The comparison between families is a whole-method comparison: besides architecture, the baselines differ from SmallFlood-CDNet in loss coefficients, initialization scheme and training script.

3.3 Training

All runs use AdamW with weight decay 10^{-4} , gradient clipping at 1.0, batch size 8, 15 epochs, 32-bit precision and no data augmentation. Training is fully deterministic: the same seed reproduces the same weights. Within a seed, variants A–D start from the same initial weights (apart from the boundary head) and see the same batch order, so differences between them are paired.

We use two recipes. In the *original* recipe, SmallFlood-CDNet and FC-Siam-Diff use a learning rate of 3×10^{-4} and BIT-SAR v2 uses 10^{-4} . In recipe *R1* all models use 10^{-4} . R1 was selected for variant A in a separate screening study that compared two learning rates and two schedules on an internal development split of training events only, without loading the validation events.

Table 2. Stages of the study. Each stage used seeds with no earlier results.

Stage	Recipe	Seeds	Evaluated on
Discovery	original	42	validation
Confirmation	original	1337, 2026	validation
Recipe screening (variant A)	four candidates	101–103	internal development split
Recipe confirmation	R1	201–203	validation
Sealed test, all 36 runs	both	all six	test, once

3.4 Metrics

The primary metric is *event-macro F1*: pixel counts are pooled within each event and the per-event F1 scores are averaged. We also report pooled pixel F1; component F1 with one-to-one matching at intersection-over-union ≥ 0.1 ; recall of small components, interior and clipped; boundary F1 with a two-pixel tolerance; and the intersection-over-union of a two-pixel inner boundary band. The decision threshold is fixed at 0.5. Results are patch-level; scenes are not reassembled.

The primary checkpoint is the last one (epoch 15). Selecting the best validation epoch chose transient early peaks in several runs, so best-epoch results are kept only as secondary information.

3.5 Staged Protocol and Decision Rules

The study proceeded in stages (Table 2). The discovery stage was exploratory. From the confirmation stage onward, each stage had a written protocol fixed before any run of that stage. Stage outcomes were judged by sign-based rules over seeds:

- *Size weighting does not improve interior-small recall*: supported if $B - A \leq 0$ and $D - C \leq 0$ in every seed, contradicted if both are positive in every seed, otherwise inconsistent.
- *The boundary package gives no inner-band gain*: the same with $C - A$ and $D - B$.
- *FC-Siam-Diff is not worse than the heavier models*: supported if FC-Siam-Diff is at least as good as A and as BIT-SAR v2 in event-macro F1 in every seed, contradicted if it is worse than the same comparator in every seed, otherwise inconsistent.

No statistical test is applied to pixels, patches or components, because they are not independent replicates.

Three disclosures apply. First, the sign rules were written after the discovery seed had been seen and were applied only to later seeds. For the original-recipe confirmation they used the best checkpoint for two of the three rules and only the two confirmation seeds; all later stages use the last checkpoint and three seeds. Second, the test evaluation was specified in full, including the list of 72

Table 3. Rule outcomes and the range of per-seed contrasts in percentage points. Size: interior-small recall, $B - A$ and $D - C$. Boundary: inner-band IoU, $C - A$ and $D - B$. Baseline: event-macro F1, FC-Siam-Diff minus A and minus BIT-SAR v2. Validation with the original recipe uses two confirmation seeds (Sect. 3.5).

Setting	Size weighting	Boundary package	FC not worse (vs. A; vs. BIT)
Validation, original	inconsistent [−10.0, +7.0]	inconsistent [−3.2, +2.1]	inconsistent [+2.7, +5.5]; [−5.6, +2.8]
Validation, R1	inconsistent [−1.1, +10.5]	inconsistent [−0.4, +6.7]	inconsistent [−3.0, +24.4]; [−27.8, +2.0]
Test, original	inconsistent [−0.4, +2.3]	inconsistent [−1.1, +2.2]	contradicted [+6.5, +13.0]; [−4.0, −2.2]
Test, R1	inconsistent [−0.1, +1.8]	inconsistent [−0.4, +0.9]	contradicted [+1.0, +14.1]; [−14.3, −3.8]

checkpoints (last and best of 36 runs), before any test array was opened, and it was run once. Third, input normalization statistics were fitted on all training events, including the two development events used for recipe screening.

3.6 Cost Measurement

Following Dehghani et al. [9] we report several indicators for one pre/post pair of 256×256 patches: parameters; FLOPs counted for convolutions and matrix multiplications only; serialized weight size; peak GPU memory; batch-1 latency in PyTorch on an NVIDIA RTX A6000 at 32-bit and 16-bit precision (100 warm-up and 500 timed runs, three repetitions); and batch-1 latency in ONNX Runtime on an Intel Xeon Silver 4310 with one and four threads (20 warm-up and 100 timed runs, three repetitions). The server was shared. GPU repetitions ran only when no other process used the GPU, apart from an idle process holding 314 MB at 0% utilization during our measurements. We did not measure an edge device, and we make no claim about one.

4 Results

4.1 The Two Components Have No Consistent Effect

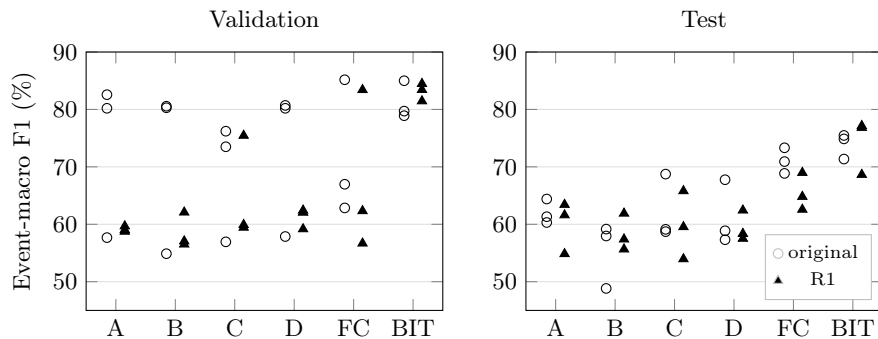
Table 3 gives the outcome of the three rules on validation and on test for both recipes. Both component rules are inconsistent in all four settings. On test the contrasts are small: between -0.4 and $+2.3$ pp for size weighting and between -1.1 and $+2.2$ pp for the boundary package, at an interior-small recall of 2–7% (Table 4).

The baseline rule is contradicted on test because BIT-SAR v2 exceeds FC-Siam-Diff in every seed of both recipes. FC-Siam-Diff in turn exceeds variant A in all six seeds.

The most direct evidence is in Table 4: with 4,899 interior small components in the test set, every model recalls between 2.4% and 6.7% of them on average.

Table 4. Test results at the last checkpoint: mean over three seeds, with the range (max – min) in brackets, in percent. “Int. small” is interior-small recall.

Recipe	Model	Event F1	Pixel F1	Comp. F1	Int. small	Bound. F1
Original	A	62.0 (4.1)	49.9 (7.8)	19.3 (1.9)	3.0 (1.0)	40.1 (2.3)
	B	55.3 (10.3)	46.3 (6.3)	19.1 (1.0)	3.2 (1.1)	37.4 (1.2)
	C	62.2 (10.0)	55.6 (13.9)	18.9 (4.8)	2.7 (1.8)	40.0 (9.8)
	D	61.3 (10.5)	52.8 (13.6)	19.2 (1.9)	3.4 (1.6)	39.2 (6.7)
	FC-Siam-Diff	71.0 (4.5)	61.1 (15.3)	19.2 (4.1)	6.7 (0.7)	50.2 (5.8)
	BIT-SAR v2	73.9 (4.1)	70.0 (4.2)	27.1 (4.9)	5.1 (2.9)	52.1 (6.0)
R1	A	60.0 (8.6)	45.6 (6.0)	17.9 (2.7)	2.4 (1.1)	37.9 (7.2)
	B	58.3 (6.2)	44.1 (6.8)	18.9 (0.9)	3.5 (0.2)	38.1 (2.3)
	C	59.8 (11.9)	47.7 (11.0)	18.0 (4.3)	2.7 (2.8)	37.6 (10.1)
	D	59.4 (4.9)	48.1 (3.8)	19.1 (1.9)	3.5 (1.8)	38.7 (2.9)
	FC-Siam-Diff	65.5 (6.4)	59.5 (11.8)	15.9 (4.4)	4.7 (4.4)	44.7 (7.6)
	BIT-SAR v2	74.2 (8.5)	70.5 (9.2)	25.6 (10.2)	4.0 (4.1)	51.4 (13.6)

**Fig. 1.** Event-macro F1 of every run at the last checkpoint. Each marker is one seed (circles: original recipe; triangles: R1). FC: FC-Siam-Diff; BIT: BIT-SAR v2.

Size weighting does not change this. The small-flood problem that motivated the design is, on this benchmark, unsolved by all six models.

4.2 Seed Sensitivity and Rank Stability

Figure 1 shows event-macro F1 for every run. On validation, the spread within one model and recipe reaches 26.7 pp (FC-Siam-Diff under R1: 62.3, 83.4 and 56.7) and 16.0 pp for variant C. The size-weighting contrast $B - A$ on interior-small recall, measured at the best checkpoint, was -13.2 pp in the discovery seed and $+1.6$ and $+7.0$ pp in the two confirmation seeds: a different sign in the seed that prompted the hypothesis than in the seeds that tested it.

On test we counted, for each of the 15 pairs of models, whether their order by event-macro F1 is the same in all three seeds of a recipe. Ten pairs are stable

under the original recipe and nine under R1. The pattern is clear. The order *between* families is stable: BIT-SAR v2 is above FC-Siam-Diff in all six seeds, and both baselines are above each SmallFlood-CDNet variant with one exception (variant C exceeds FC-Siam-Diff in one R1 seed, 65.8 against 64.8). The order *among* variants A–D is stable for only one pair under each recipe. A study that ranked A–D from one seed would have reported an ordering that the next seed reverses.

4.3 Variation Concentrates in One Event, but Flood Fraction Does Not Explain It

On validation, almost all of the seed-to-seed spread comes from one event. Coraki and Hebei are comparatively stable for every model (F1 of about 90–95% and 68–86% under R1), whereas Lumberton F1 ranges from below 10% to above 80%. Under R1, variant A is below 20% F1 on Lumberton in 42 of its 45 training epochs across the three seeds. The failure is over-prediction: in epochs 11–15 the SmallFlood-CDNet variants predict 1.5–4.4% of pixels as flood where the truth is 1.08%, while training loss is almost identical across seeds. BIT-SAR v2 never falls below 20% on Lumberton in any of its 90 epochs across the six seeds.

Lumberton has a low flood fraction (about 0.44% of valid pixels, against 1.08% for the validation set as a whole), so we wrote down a rule before opening the test set: the finding *generalizes* if the test event with the lowest flood fraction has the largest across-seed F1 range in at least 7 of the 12 model-and-recipe groups, and *does not generalize* if in at most 4. The low-fraction test event is Jubba, and it has the largest range in 4 groups. The outcome is “does not generalize”.

Variation on test is still concentrated in one event, but it is Nova Kakhovka, the event with the *highest* flood fraction and almost all of the small components. It has the largest range in the other 8 groups. All 14 runs with a test event below 20% F1 have Nova Kakhovka as that event, all 14 are SmallFlood-CDNet variants, and in 12 of them the model predicts more flood pixels than the truth. No FC-Siam-Diff or BIT-SAR v2 run falls below 20% on any test event. We can therefore say that seed sensitivity is event-specific, and that it took the form of over-prediction on one event per split for the smallest model family. We cannot say which property of an event causes it; flood fraction is not it.

4.4 A Recipe Chosen on Held-Out Training Events Did Not Transfer

Because the original recipe collapsed on Lumberton in one of three seeds, we screened four recipes for variant A on two training events held out as a development split. The lower learning rate with a constant schedule (R1) gave the highest and most balanced development score in all three seeds (mean event-macro F1 50.2% against 37.4% for the original recipe) and met a stability rule fixed in advance.

Table 5. Cost for one pair of 256×256 patches. FLOPs count convolutions and matrix multiplications only. GPU: PyTorch on an RTX A6000, batch 1, median latency. CPU: ONNX Runtime on a Xeon Silver 4310, batch 1, median latency. Size and peak memory are for 32-bit weights.

Model	Params	FLOPs	Size	GPU (ms)		CPU (ms)		Mem.
	(M)	(G)	(MB)	FP32	FP16	1 thr.	4 thr.	(MB)
SmallFlood-CDNet (A, B)	1.08	0.35	4.5	8.7	10.5	10.7	7.5	10.0
+ boundary head (C, D)	1.08	0.35	4.5	9.2	10.8	10.6	8.6	10.0
FC-Siam-Diff	0.49	7.02	2.0	3.0	3.8	81.4	26.0	56.0
BIT-SAR v2	3.49	21.43	14.1	24.5	25.7	350.9	163.3	83.4

On the real validation events R1 did not help. Variant A reached 58.7, 59.0 and 59.7% event-macro F1 in its three R1 seeds, against 82.6, 80.2 and 57.7% for three other seeds under the original recipe, with Lumberton F1 of 8.8, 3.8 and 6.1%. The comparison is not paired, since the seeds differ, but the direction is plain: the lower learning rate made the failure consistent instead of removing it. On test the two recipes are close for variant A (62.0% and 60.0%), R1 is worse for FC-Siam-Diff (71.0% and 65.5%), and BIT-SAR v2, which used the same learning rate in both, is unchanged (73.9% and 74.2%).

This stage also exposed a weakness in our own stability rule. It required a small range of event-macro F1 across seeds and a per-seed F1 on the hardest event of at least half of the median over seeds. A model that fails in every seed has a low median and passes. Variant A under R1 was therefore labelled “stable” at a Lumberton F1 below 10%. We noticed this before the last seed finished, kept the rule as written, and report the label together with its meaning: consistent, not good. A stability rule needs an absolute floor or a reference model.

4.5 Accuracy and Cost

Table 5 shows that the four usual indicators rank the models differently. By parameters, FC-Siam-Diff is the smallest. By counted FLOPs, SmallFlood-CDNet is lightest by a factor of 20 over FC-Siam-Diff and 62 over BIT-SAR v2. By GPU latency, FC-Siam-Diff is fastest and SmallFlood-CDNet is 2.9 times slower despite its low FLOP count. By CPU latency, SmallFlood-CDNet is fastest: on one thread FC-Siam-Diff is 7.6 times and BIT-SAR v2 32.8 times slower. This is the pattern that Dehghani et al. [9] warn about, observed here within a single task.

Half precision did not reduce batch-1 latency on this GPU for any model, although it roughly halved the stored size and reduced peak memory. On the validation split, 16-bit inference agreed with 32-bit inference on at least 99.989% of valid pixels and changed no reviewed metric by more than 0.18 pp; ONNX Runtime agreed with PyTorch on all pixels of the 64 patches we compared. Timing on a shared server is noisy, and some repetitions differed visibly; we report the values as measured.

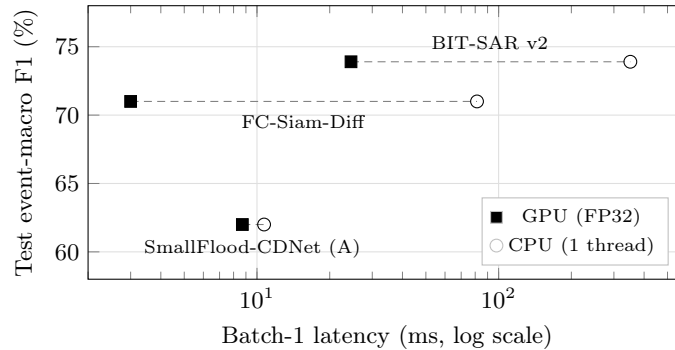


Fig. 2. Accuracy against latency. Test event-macro F1 is the mean over three seeds under the original recipe (Table 4); latency is from Table 5. Each model appears twice, once per device, joined by a dashed line.

Figure 2 combines Tables 4 and 5. The model that is cheapest on a CPU is the least accurate, and the model that is most accurate on test, and the only one that never collapsed on a validation event, is the most expensive on every indicator. FC-Siam-Diff sits between them, with the caveat that its accuracy depends more on the recipe.

5 Discussion

What the negative result means. The component effects that looked interpretable in one seed are within the variation between seeds. The discovery seed suggested that size weighting *reduced* interior-small recall by 13 points; later seeds showed changes of either sign, and on a test set with thirteen times more small components the contrasts are within about two points. We do not conclude that size weighting or boundary supervision can never help. We conclude that, for this model, dataset and training budget, a benefit could not be shown, and that a single-seed study would have reported one effect or its opposite depending on the seed.

What is not established. We do not know why one event per split triggers over-prediction in the smallest model family. We tested and rejected one explanation, flood fraction. We also do not know why BIT-SAR v2 is robust. It differs from the other models in architecture, initialization and loss coefficients at once, so its advantage cannot be attributed to any one of them. Separating these factors is a natural next study.

Recommendations. For studies of this kind we suggest: (i) report several seeds with the range, not one run; (ii) report per-event results, because an event-macro score over three events can be decided by one of them; (iii) report the

last checkpoint next to the best one; (iv) report object-level metrics with their denominators, since pixel F1 hides a 2–7% small-region recall; (v) report several cost indicators on more than one device; and (vi) write the comparison rule down before looking, and say when it was written.

Limitations. The study uses one dataset, three validation events and three test events, and three seeds per recipe; the sign rules are conventions, not statistical tests. Inputs are intensity only, although coherence is known to help in urban areas [14]. Evaluation is patch-level, training lasts 15 epochs without augmentation, and the split is our own. The recipe was tuned for one variant only. The sign rules were written after a discovery seed, and one stability rule had the loophole described in Sect. 4.4. Cost was measured on a shared server and not on edge hardware, and the FLOP counter ignores normalization, activation and attention-softmax operations. Our literature search was bounded.

6 Conclusion

We set out to improve the detection of small urban floods in SAR imagery with a lightweight model, and instead documented three things. First, neither a size-weighted loss nor a boundary package had a consistent effect across six seeds, two recipes and a sealed test set, and all models recovered only a few percent of interior small floods. Second, rankings between model families were stable while rankings among close variants were not, with the variation concentrated in one event per split for reasons that flood fraction does not explain. Third, parameters, FLOPs, GPU latency and CPU latency ordered the models differently, so “lightweight” needs a device and an indicator attached to it.

The open questions are specific: what makes the BIT-based model robust, whether the findings hold on a second SAR flood dataset, whether coherence changes the picture for urban floods, and what the models cost on an actual edge device.

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References

1. Bonafilia, D., Tellman, B., Anderson, T., Issenberg, E.: Sen1Floods11: A georeferenced dataset to train and test deep learning flood algorithms for Sentinel-1. In: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops (2020)

2. Bountos, N.I., et al.: Kuro Siwo: 33 billion m² under the water. a global multi-temporal satellite dataset for rapid flood mapping. In: *Advances in Neural Information Processing Systems (Datasets and Benchmarks Track)*. vol. 37 (2024)
3. Bouthillier, X., Delaunay, P., Bronzi, M., Trofimov, A., Nichyporuk, B., Szeto, J., Sepah, N., Raff, E., Madan, K., Voleti, V., Kahou, S.E., Michalski, V., Serdyuk, D., Arbel, T., Pal, C., Varoquaux, G., Vincent, P.: Accounting for variance in machine learning benchmarks. In: *Proceedings of Machine Learning and Systems (MLSys)*. vol. 3 (2021)
4. Cao, Z., Wu, M., Yan, R., Zhang, F., Wan, X.: Detection of small changed regions in remote sensing imagery using convolutional neural network. *IOP Conference Series: Earth and Environmental Science* **502**(1), 012017 (2020). <https://doi.org/10.1088/1755-1315/502/1/012017>
5. Chen, H., Qi, Z., Shi, Z.: Remote sensing image change detection with transformers. *IEEE Transactions on Geoscience and Remote Sensing* **60** (2022). <https://doi.org/10.1109/TGRS.2021.3095166>
6. Codegoni, A., Lombardi, G., Ferrari, A.: TINYCD: A (not so) deep learning model for change detection (2022), arXiv:2207.13159
7. Corley, I., Robinson, C., Ortiz, A.: A change detection reality check (2024), arXiv:2402.06994
8. Daudt, R.C., Le Saux, B., Boulch, A.: Fully convolutional Siamese networks for change detection. In: *IEEE International Conference on Image Processing (ICIP)* (2018)
9. Dehghani, M., Arnab, A., Beyer, L., Vaswani, A., Tay, Y.: The efficiency misnomer. In: *International Conference on Learning Representations (ICLR)* (2022)
10. Howard, A., Sandler, M., Chu, G., Chen, L.C., Chen, B., Tan, M., Wang, W., Zhu, Y., Pang, R., Vasudevan, V., Le, Q.V., Adam, H.: Searching for MobileNetV3. In: *IEEE/CVF International Conference on Computer Vision (ICCV)* (2019)
11. Kaçmaz, A., Alganci, U.: SAR-based flood extent mapping with a lightweight Siamese U-Net and differential attention mechanism. *Earth* **7**(3), 87 (2026). <https://doi.org/10.3390/earth7030087>
12. Kervadec, H., Bouchtiba, J., Desrosiers, C., Granger, E., Dolz, J., Ben Ayed, I.: Boundary loss for highly unbalanced segmentation. In: *Medical Imaging with Deep Learning (MIDL), Proceedings of Machine Learning Research*. vol. 102 (2019)
13. Kofler, F., et al.: blob loss: Instance imbalance aware loss functions for semantic segmentation (2022), arXiv:2205.08209
14. Li, Y., Martinis, S., Wieland, M.: Urban flood mapping with an active self-learning convolutional neural network based on TerraSAR-X intensity and interferometric coherence. *ISPRS Journal of Photogrammetry and Remote Sensing* **152**, 178–191 (2019)
15. Maier-Hein, L., Reinke, A., et al.: Metrics reloaded: Recommendations for image analysis validation. *Nature Methods* **21**(2), 195–212 (2024). <https://doi.org/10.1038/s41592-023-02151-z>
16. Mateo-Garcia, G., Veitch-Michaelis, J., Smith, L., et al.: Towards global flood mapping onboard low cost satellites with machine learning. *Scientific Reports* **11** (2021). <https://doi.org/10.1038/s41598-021-86650-z>
17. Picard, D.: Torch.manual_seed(3407) is all you need: On the influence of random seeds in deep learning architectures for computer vision (2021), arXiv:2109.08203
18. Rachmadi, M.F., Poon, C., Skibbe, H.: Improving segmentation of objects with varying sizes in biomedical images using instance-wise and center-of-instance segmentation loss function. In: *Medical Imaging with Deep Learning (MIDL), Proceedings of Machine Learning Research*. vol. 227, pp. 286–300 (2024)

19. Reimers, N., Gurevych, I.: Reporting score distributions makes a difference: Performance study of LSTM-networks for sequence tagging. In: Proceedings of the Conference on Empirical Methods in Natural Language Processing (EMNLP). pp. 338–348 (2017). <https://doi.org/10.18653/v1/D17-1035>
20. Saleh, T., Holail, S., Xiao, X., Xia, G.S.: High-precision flood detection and mapping via multi-temporal SAR change analysis with semantic token-based transformer. *International Journal of Applied Earth Observation and Geoinformation* **131**, 103991 (2024). <https://doi.org/10.1016/j.jag.2024.103991>
21. Saleh, T., Weng, X., Holail, S., Hao, C., Xia, G.S.: DAM-Net: Flood detection from SAR imagery using differential attention metric-based vision transformers. *ISPRS Journal of Photogrammetry and Remote Sensing* (2024), arXiv:2306.00704
22. Salehi, S.S.M., Erdogmus, D., Gholipour, A.: Tversky loss function for image segmentation using 3D fully convolutional deep networks. In: *Machine Learning in Medical Imaging (MLMI)* (2017)
23. Tomanič, T., Baudhuin, A., Sotošek, J., Brence, J., Panov, P., Simidjievski, N., Kocev, D.: A comprehensive and trustworthy benchmark of AI methods for change detection in Earth observation (2026), arXiv:2608.28247
24. Zhao, J., Xiong, Z., Zhu, X.X.: UrbanSARFloods: Sentinel-1 SLC-based benchmark dataset for urban and open-area flood mapping. In: *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops (EarthVision)* (2024), arXiv:2406.04111